

Acoustic-Video Perception for Robot Manipulation with Occluded Contact Dynamics

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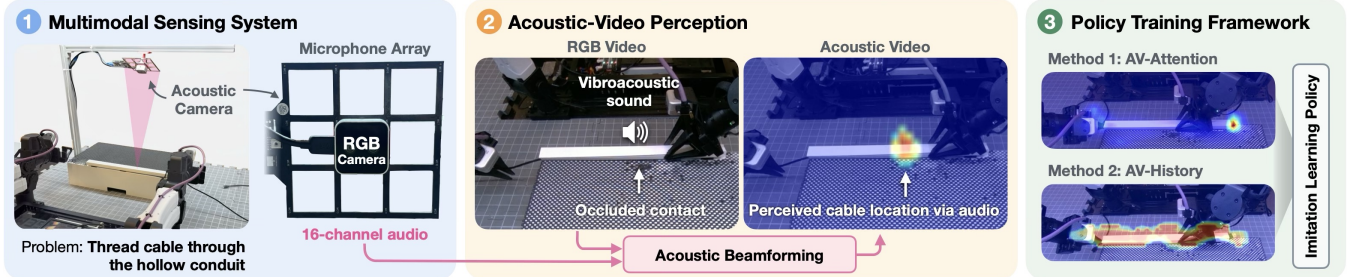


Fig. 1. Overview of the Acoustic Video (AV) robot perception system. The overhead acoustic camera localizes the sound source on the RGB camera plane. We integrate this modality into visuomotor policies in two ways: (1) AV-guided attention supervision and (2) AV-history auxiliary observation.

Abstract—Almost all manipulation tasks involve occluded contact dynamics: it is inherently difficult to visually observe detailed surface contacts as they are created and changed. Consider grasping, assembly, or pushing – even a planar task like “Push T” hides the contact between object and surface. In some cases, occluded contact generates perceptible *vibroacoustic* sound through friction and provides useful contact information. In this paper, we explore how a low-cost “acoustic camera” can perceive occluded contact dynamics to enhance robot manipulation. We use microphone array beamforming to localize sound over time onto the 2D image plane of an RGB camera, producing a spatially aligned representation we call *Acoustic Video*. We present two approaches, AV-Attention and AV-History, for training manipulation policies with Acoustic Video using an encoder-only Action Chunking Transformer. Across five contact-rich manipulation tasks, experiments suggest that acoustic video can improve task success rate by 21.0% and progress by 16.1% on average compared to vision-only baseline and conventional mono-channel audio approach. Video figures are available at <https://rgba-robot.github.io/>.

I. INTRODUCTION

Humans and many animals use spatial audio perception [1]. Inspired by nature, audio has long been used in robotics: as early as 1989, a small autonomous robot called Squirt [2] used sound to guide its movement, mimicking the behavior of a bug that ventures out in the direction of noises. Microphone arrays¹ were developed in the early 20th century during World War I, originally to triangulate gun positions based on sound arrival times [3]. Since then, microphone arrays have also been used in robotics, such as drones for search-and-rescue tasks [4], [5], human-robot interaction (HRI) for conversational agents [6], [7], and manipulation using contact microphones [8]–[11]. Despite this history, using spatial acoustics directly grounded in the visual scene for a robot manipulation policy is underexplored.

¹Paper under review.

¹a set of two or more individual microphones placed in a specific geometric layout to capture sound from multiple points in space at the same time.

Many manipulation tasks involve occluded contact dynamics that are difficult to observe visually as they form and change. Consider threading a cable through a hollow conduit (Fig. 1), where the occluded endpoint location is critical for identifying the task stage. Even a planar task like “Push-T” depends on surface contact underneath the object, and the correct pushing direction depends on how contact evolves. These interactions, however, make friction between the contacting surfaces and radiate *vibroacoustic*² sound. This sound provides a useful contact cue, but prior approaches perceive it with a single-channel signal that carries no spatial information about the contact [12]–[19]. As a result, the policy must implicitly learn the relationship between acoustic and visual observations from inherently limited information.

This paper explores acoustic video perception for tasks involving *vibroacoustic* sound. The array microphone (*Acoustic Camera* [20]) can localize the sound in the scene during manipulation. We use acoustic beamforming at the target frequency associated with contact friction (10 kHz). We then create time-varying acoustic intensity maps overlaid on an RGB video stream, which we call *Acoustic Video* (AV).

We propose two methods to integrate this novel modality into imitation learning policies: 1) **AV-Attention** supervises the model’s visual attention mechanisms to focus computation on active contact regions. 2) **AV-History** incorporates acoustic video as a policy observation to track task progress over time. While the AV-History method uses live audio as a reactive input at deployment, the AV-Attention method acts purely as a training-time framework, allowing inference on standard vision inputs.

We evaluate the potential of our Acoustic Video perception with these manipulation tasks as shown in Fig. 4: thread cable through a conduit, grind with a rotary tool, peel transparent film on smartphone, cut seal to decant box,

²sound radiated by a vibrating structure, here induced by friction at the contact.

and wipe blue line from product. Evaluations on five real vibroacoustic tasks suggest that acoustic-video perception can improve task success rate by up to 21.0% and task progress by 16.1% compared to models trained on vision only and a conventional mono-channel audio approach.

II. RELATED WORK

We first review prior work on multimodal perception for manipulation, and then discuss two topics more specific to our approach, spatial audio in robotics, and spatial or auxiliary supervision for robot learning.

A. Multimodal Perception for Robot Manipulation

In contact-rich manipulation, interaction state is often only partially observable from vision: contact, friction, slip, and applied force are hard to infer visually, especially under occlusion or on transparent and specular objects. Prior work has therefore paired vision with force [21]–[23], tactile [24]–[27], acoustic [12], [18], and capacitive [28]–[30] sensing, as well as imaging beyond the visible spectrum [31], [32].

Acoustic sensing is promising because microphones are inexpensive and capture contact sound remotely, through air or a solid medium, beyond the region the sensor touches. Some approaches actively excite the object and measure its acoustic response [10], [33], but many tasks produce sound passively: tapping, wiping, and pushing all excite micro-friction vibrations that radiate as *vibroacoustic* sound. Prior work exploits this signal to manipulate under occlusion, but mostly consumes mono-channel audio as waveforms or spectrograms [12]–[19]. While these works show that sound reveals task-relevant information hard to infer from vision alone, mono-channel audio carries no spatial information. Yet in many tasks, such as threading a cable (Fig. 1), the contact location is exactly the cue the policy needs, so it must implicitly learn to bind the sound to a place in the visual scene, a mapping that is costly and sometimes infeasible to learn.

Our work is similarly motivated but uses remote rather than contact-mounted acoustic sensing. A microphone array observes interaction sounds without instrumenting the object or contact surface, while spatially grounding the signal in the visual scene. This enables fusing acoustic and visual features at corresponding image locations, which prior work leaves underexploited.

B. Spatial Audio in Robotics

Humans and many animals use spatial audio perception to understand the world [1]. In robotics, spatial audio has been used in three main ways: sonar ranging, sound source separation, and sound source localization.

Sonar has a long history in mobile robotics, where early systems measured proximity to obstacles by emitting a pulse and timing the return echo [34], [35]. Unlike LiDAR, sonar reflects off transparent surfaces, and recent work fuses it with depth sensing to reconstruct glass walls via a learned implicit representation [36].

Closer to our work, microphone arrays have been used to separate a target sound from noise. This line of research, termed robot audition [6], [37], was motivated by human-robot interaction, where a robot must isolate a speaker from competing talkers or its own motor noise. Beamforming toward a known source direction enabled real-time separation of simultaneous speakers [38], and open-source robot audition packages have since emerged [7], [39]. Commercial humanoids such as the Unitree G1 and R1 now ship with built-in arrays for voice interaction.

Most related to our work, microphone arrays have been used to localize sound sources: an eight-microphone array let a mobile robot follow moving speakers [40], and array-equipped drones have localized victims for search-and-rescue where vision is obstructed [4], [5]. In manipulation, a few works use contact microphones on the end effector or arm to localize collisions [8], [9], [11]. These works, however, either reduce the source to a direction of arrival or are bounded to the robot’s own body; none localizes sound in the visual task scene for manipulation. In contrast, we ground acoustic information densely in the visual scene, so that a learning-based policy can directly exploit where contact occurs.

C. Spatial and Auxiliary Supervision for Robot Learning

Robot policies are commonly trained end-to-end from demonstrations [41] or task rewards, but a growing body of work shows that explicitly providing task-relevant spatial structure improves policy learning and generalization. Human gaze has been used to direct visuomotor representations toward manipulation-relevant regions [42], [43] and, more recently, as auxiliary regularization for imitation learning and vision-language-action policies [44]–[46]. Beyond gaze, spatial supervision has been introduced through language-conditioned keypoints [47], pixel-aligned contact representations [48], affordance predictions [49], and auxiliary objectives on object grounding, spatial geometry, and temporal task structure [50]. Our work similarly supervises policy attention with explicit spatial structure, but derives that structure directly from physical interaction rather than from human gaze, manual labels, or visually inferred affordances: we localize interaction sounds in the camera frame and use the resulting acoustic video as a physically grounded signal of where task-relevant interaction occurs.

III. PROBLEM SETUP

We consider contact-rich manipulation tasks that involves vibroacoustic sound, the passive sound by contact friction. Our goal is to learn a visuo-acoustic policy from demonstrations that exploits this sound by localizing it in the image frame that explicitly aligns the acoustic signal with the corresponding visual features. We assume a static environment in which the manipulated object does not substantially shift its position relative to the acoustic camera. We also assume a relatively consistent acoustic background (e.g., HVAC or AC-driven lighting) without abrupt or transient noise.

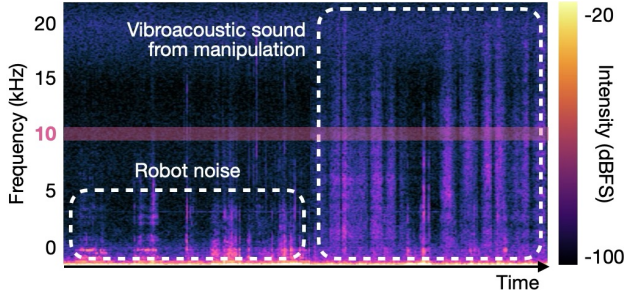


Fig. 2. Overall acoustic profile from manipulation vs. robot noise. The frequency 10 kHz is used for acoustic beamforming.

IV. ACOUSTIC-VIDEO PERCEPTION SYSTEM

A. Acoustic Camera Hardware

To capture spatially grounded visuo-acoustic observations, we implemented an overhead acoustic camera with two commercial components: a 16-channel microphone array (miniDSP UMA-16 v2) and an RGB camera (Intel RealSense D405 camera). The camera is mounted in the center of the array, as shown in Fig. 1. The microphone streams synchronized 16-channel audio sampled at 44.1 kHz over a USB 3.0 interface. The RGB camera records an overhead image at a resolution of 1280×720 at 30 Hz. We calibrate the array’s computed acoustic maps to match the camera’s field of view (FoV; $89.0^\circ \times 57.9^\circ$). To this end, we define the beamforming search grid as a planar region at a fixed depth D from the array, sized to span the camera’s FoV at that depth. Given FoV, the dimensions of the grid are $W = 2D \tan\left(\frac{89.0^\circ}{2}\right)$, $H = 2D \tan\left(\frac{57.9^\circ}{2}\right)$ with a 2 cm spacing, so the acoustic map is aligned with the RGB image.

B. Acoustic Beamforming

Consider the friction sound produced when the cable moves through a hollow conduit as shown in Fig. 1. This sound propagates to each of 16 microphones at slightly different times due to their spatial disparity. For every candidate location $\mathbf{s}_{xy}$ on the planar search grid defined in Sec. IV-A, we first compute the distance $\|\mathbf{s}_{xy} - \mathbf{m}_i\|$ from the candidate source to each microphone $\mathbf{m}_i$. Dividing by the speed of sound c , we get the propagation delay $\frac{\|\mathbf{s}_{xy} - \mathbf{m}_i\|}{c}$. We then realign the 16-channel audio by these delays and sum them. The construction from the source would constructively interfere to amplify the signal, and others would destructively interfere and cancel out noise.

In this process, we specifically combine Multiple Signal Classification (MUSIC) [51] and Steered Response Power (SRP) [52] in a complementary way. We use SRP on broadband audio for a robust yet spatially diffuse sound-pressure map. We then use MUSIC at the target frequency for tight sound localization.

As shown in Fig. 2, contact dynamics during manipulation produce high-frequency acoustic signatures, exceeding 10 kHz. This distinct frequency profile enables separation of task-relevant contact audio from low-frequency robot motor noise (0–4 kHz) and mechanical vibrations or ambient (roomtone) sound (< 100 Hz). We therefore high-pass the raw signal at 3 kHz before feeding the broadband audio into

SRP. For MUSIC, we select a single narrowband target at 10 kHz with a 517 Hz bandwidth. In principle, the MUSIC can use multiple target frequencies or change them adaptively, but we keep it fixed to minimize computation for real-time throughput. We threshold the SRP map at the 90-th percentile to form a spatial gate and multiply it element-wise by the MUSIC map to obtain the acoustic map, $A(x, y, t)$.

C. Acoustic-Video Perception Input

We define two temporally accumulated observations of sound-pressure maps: History and Future representations.

In the History representation $H(x, y, T)$, we max-pool all past acoustic maps up to elapsed frame T to track cumulative task progression, i.e., $H(x, y, T) = \max_{0 \leq t \leq T} A(x, y, t)$. Semantically, it is the surface areas that have been cleaned in Fig. 3a). This representation is provided as a policy observation in Method 2, AV-History.

The Future representation $F(x, y, T)$ is the opposite of it – it max-pools remaining acoustic signals from frame T onward, i.e., $F(x, y, T) = \max_{t \leq T \leq T_{final}} A(x, y, t)$, where T is the elapsed time and T_{final} is the demonstration time. Semantically, it is the remaining target areas to be further cleaned in Fig. 3a). This is computed offline only in training demo episodes. This target supervises the visual attention maps in Method 1, AV-Attention.

V. ACOUSTIC-VIDEO POLICY LEARNING FRAMEWORK

We propose two policy learning frameworks to use AV perception: AV-Attention and AV-History. We first describe the base visuomotor policy architecture that both frameworks share, then explain each framework on this base architecture.

A. Base Visuomotor Policy Architecture

To use AV for robot control, we consider the Action Chunking Transformer (ACT) [53] with an encoder-only design for its efficient performance on a small amount of demonstrations. The gray-colored components in Fig. 3b illustrate the base policy architecture. We replaced ACT’s ResNet vision encoder with a frozen DINOv2 with registers [54] to better interpret the effects of attention supervision from Method 1 without high-norm artifact patches. All camera frames are encoded by this vision encoder, yielding 700 patch tokens (20×35) for the overhead view and 140 (10×14) for each wrist view. The 14-DoF bimanual joint configuration is embedded into a single proprioceptive token by an MLP, and the action chunk is represented by 50 learned action queries. We concatenate all these tokens in a single sequence (1031 length $\times$ 384 dimension) and process it with one stack of 3 self-attention blocks with 12 heads each. We intentionally make our architecture shallow to avoid overfitting on the limited demonstration data while retaining enough attention heads to fuse the multi-view and AV tokens. We use the same action chunk size (1-second) as the original ACT; a three-layer multilayer perceptron (MLP) decodes the hidden states into a chunk of 50 joint displacements relative to input proprioception.

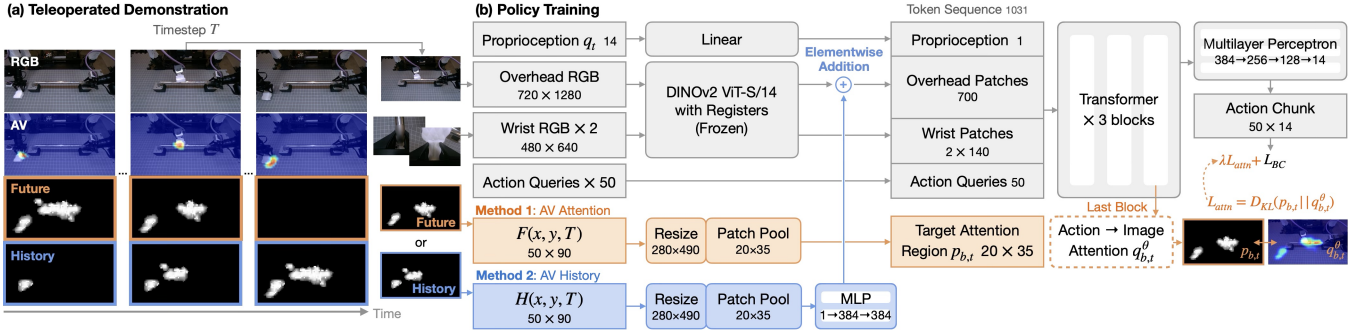


Fig. 3. Each frame of the demo episode includes three camera frames (1 overhead, 2 wrists) and an AV observation that constructs the input of each method. A path with orange color indicates method 1, **AV-Attention**: a target attention region ($p_{b,t}$) is used to compute the Kullback-Leibler (KL) divergence loss (L_{attn}) with transformer attention ($q_{b,t}^{\theta}$). Then this loss is added to the behavior cloning loss (L_{BC}) to update the model weights. Note that this attention supervision is done only on the overhead RGB frame. The path with blue indicates method 2, **AV-History**: a two-layer multilayer perceptron (MLP) embeds the per-patch AV-history scalar into a feature space, and the result is added element-wise to the overhead patches.

B. Method 1: AV-Attention

In our first method, we use acoustic-video perception to guide the imitation learning policy to attend to visual features directly related to the manipulation task. Following the convention of GuidedVLA [50] and Gaze-Regularized VLA [45], we add a training-only regularization term to the original behavior cloning loss. The orange-colored components in Fig. 3 show the implementation of this. Specifically, we update the model weights through the loss:

$$\mathcal{L}(t) = \mathcal{L}_{BC}(t) + \lambda \mathcal{L}_{attn}(t), \quad \lambda = 0.05 \quad (1)$$

$$\mathcal{L}_{BC}(t) = \|a_{gt}^{(t)} - a_{\theta}^{(t)}\|_1 \quad (2)$$

$$\mathcal{L}_{attn}(t) = \frac{1}{|\mathcal{G}_t|} \sum_{(b,t) \in \mathcal{G}} D_{KL}(p_{b,t} \| q_{b,t}^{\theta}) \quad (3)$$

where $p_{b,t}$ is the future-accumulated sound-pressure map (Fig. 3a Future) pooled onto the patch grid, and $q_{b,t}^{\theta}$ is the attention from the action query to the image patches in the final transformer block, averaged over heads. $\mathcal{G}$ excludes observations whose target map is empty, i.e., the silent tail after the last acoustic event. We deliberately chose a small regularization weight of $\lambda=0.05$ per the findings of Pani and Yang [45], which describes that maintaining low regularization strength benefits from the statistical tendencies of attention supervision labels while preserving flexibility to discover task-optimal patterns that may occasionally deviate from the label. This attention supervision is applied exclusively to the overhead camera stream. While wrist views are not directly supervised, all camera views process tokens through the same shared attention heads. Crucially, Method 1 requires acoustic sensing only during training. In deployment, the policy operates on standard vision-only observations without requiring an acoustic camera or audio processing.

C. Method 2: AV-History

In this second method, we use the acoustic video as a reactive policy input during deployment. We feed the temporally accumulated “history” map, the per-pixel maximum over all frames up to the current one (Fig. 3a, History). The blue-colored components in Fig. 3 show the integration of AV-History into the policy architecture.

The 50×90 input history map is resized and average-pooled onto the same 20×35 patch grid as the overhead image, yielding one scalar per patch. The AV history map does not pass through the DINO encoder. Instead, each scalar is embedded by a lightweight per-patch, two-layer MLP. The AV history already shares the same view geometry as the overhead image. To leverage this known spatial correspondence without learning it through attention, we apply element-wise addition of AV history features to the corresponding overhead DINO token, instead of concatenating it as additional tokens. The token count therefore is identical to the Vision-Only model, which adds no additional attention cost when the acoustic-video modality is incorporated.

VI. EXPERIMENTS

To study the efficacy of our proposed perception system, we evaluated imitation learning policies on five vibroacoustic manipulation tasks in real environments, compared to two baseline policies. We ran 20 trials per task, which yielded 400 evaluation trials in total (4 policies $\times$ 5 tasks $\times$ 20 trials).

A. Vibroacoustic Manipulation Tasks

We focus on tasks involving vibroacoustic sound that is naturally induced by friction during manipulation.

Task 1: Cable Threading. The robot routes a USB-A cable through a hollow conduit until the connector exits, then grasps it to pull it through the conduit (Fig. 4 *Cable Threading*). The cable is visually occluded inside the raceway, and spatial audio provides internal contact state. A trial is considered successful if the robot pulls it successfully.

Task 2: Plastic Grinding. The robot picks up a 3D-printed part and brings it to a rotary filing tool (Dremel) to abrade a 1 cm blue mark until the majority of the mark is removed (Fig. 4 *Plastic Grinding*). Acoustic emission from the grinding contact provides an explicit contact cue. A trial is considered successful if the robot abrades the majority (i.e., 50%) of the blue mark.

Task 3: Transparent Film Peeling. The robot peels a transparent film from a smartphone screen (Fig. 4 *Transparent Film Peeling*). The policy should detect failure under manipulation of transparent material and re-attempt grasping based on spatial acoustic feedback. A trial is considered successful if the film is fully detached from the screen.

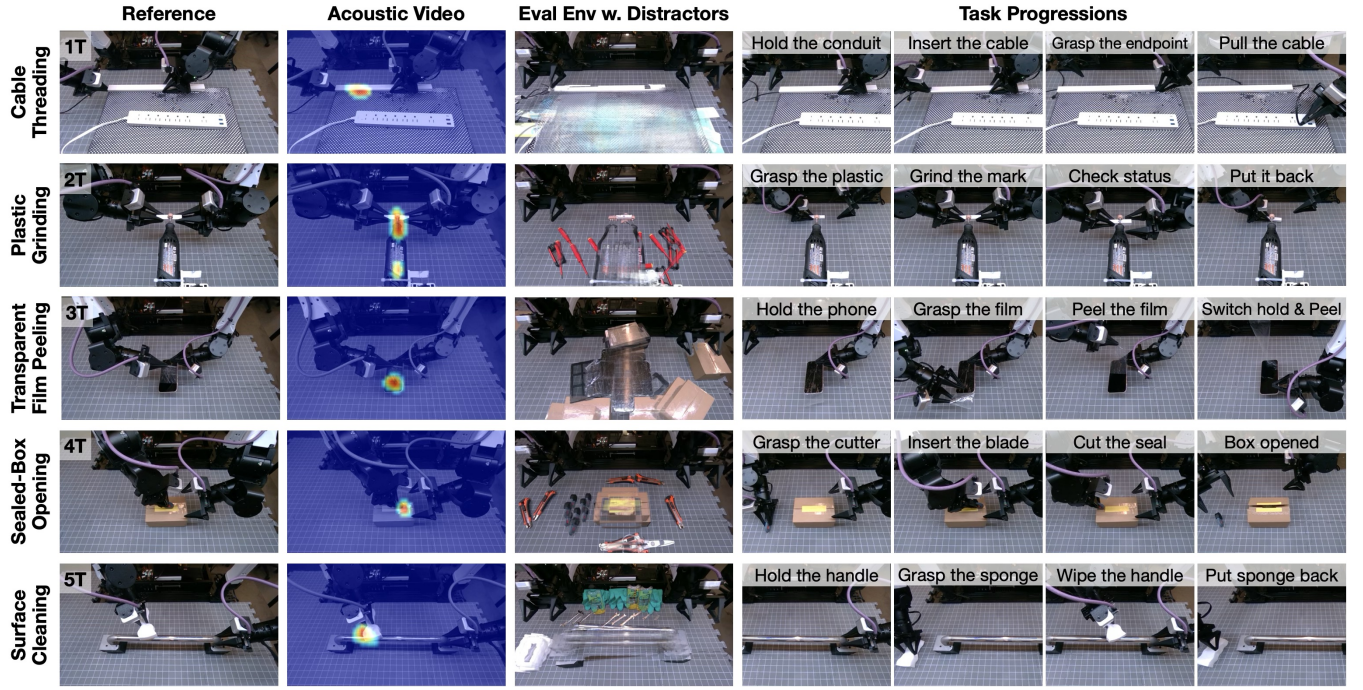


Fig. 4. Five real-world environments of our experiment with distractor objects. The distractor objects were not in the training data and were placed in physical space during rollout.

Task 4: Sealed-Box Opening. The robot grasps a utility cutter, slices through a 9 cm tape seal on a box, then the box flaps are opened (Fig. 4 *Sealed-Box Opening*). The acoustic signal confirms the cutting contact. A trial is considered successful if the tape is fully cut and the box is unsealed.

Task 5: Surface Cleaning. The robot wipes a 29 cm target mark off an industrial handle (Fig. 4 *Surface Cleaning*). Specular glare on the metallic surface impairs visual tracking. A trial is considered successful if the robot cleaned $>67\%$ of the blue mark.

B. Baselines

We evaluate 2 baseline approaches: Vision-Only and ManiWAV-Inspired. To ensure a fair comparison, all baseline policies are trained and evaluated using identical datasets, hyperparameter settings, and deployment parameters (1 second action chunk size, 20 Hz inference frequency, and exponential temporal aggregation decay of 0.05) as our proposed methods. The only differences are in acoustic-video observations and the corresponding architecture described in Fig. 3 (blue, orange). We have two baseline policies, and the description of each architecture follows:

1) Vision Only evaluates policy performance using visual observations from three camera streams (overhead and two wrist views) in Fig. 3 (gray).

2) ManiWAV Inspired adapts ManiWAV [12], which feeds a single-channel audio spectrogram to the policy. We use channel 1 (center-left) of our array, downsampled from 44.1 to 16 kHz, and extract a 64-bin log-mel spectrogram (25 ms Hann window, 10 ms hop) following [12]. At each step, a 2.0 s spectrogram history is encoded by a learned CNN into 8 audio tokens, which are concatenated with the proprioception, visual, and action-query tokens (1039 tokens

total). Following [12], we apply ESC-50 environmental audio augmentation and spectrogram dropout during training, which we found essential to prevent overfitting to the audio modality. We note that this baseline is inspired by rather than identical to ManiWAV, as we do not use a contact microphone or a diffusion UNet.

We compare these two baselines with our proposed methods, which yield four policy models for evaluation per task.

C. Policy Training & Deployment

We collected 40 demonstrations for each task using a teleoperation setup. We train policies with an action chunk size of 1.0 s on NVIDIA GeForce RTX 5090 GPU. During rollout, the policy runs inference at 20 Hz (every 50 ms) and applies temporal aggregation across overlapping action chunks using an exponential decay factor of 0.05. All model inference is executed on an Ubuntu 24.04 LTS desktop with the same GPU machine, processing real-time data streams from three cameras and the 16-channel microphone array.

D. Real Rollout Environments

For real-world evaluation, we execute 20 rollout trials per method, per task. Each trial is terminated upon successful task completion or reaching a maximum timeout (2 minutes, roughly $2\times$ of averaged teleoperated demonstration).

To understand whether acoustic-video perception helps the policy to learn task-related features better, we added visual distractors (i.e., objects that were not seen in the training dataset) around the task environment. This includes boxes, screwdrivers, cutters, gloves, and a cleaning liquid bottle, etc (zero to three distractors per scene). We also varied the task object position by 5-10 cm displacement. The 20 evaluation environments for each task are visualized in the last row of

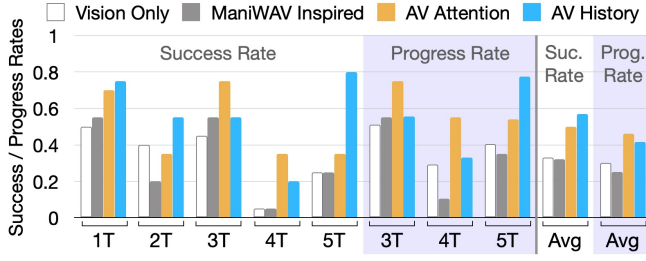


Fig. 5. End-to-end success on real tasks. Each task is evaluated on 20 trials for each policy, which yields 400 evaluation trials in total. In addition to success rate, we report progress rate for the three tasks in which continuous execution (e.g., cutting the entire seal) is required for success.

Fig. 4 *Eval Env*. For a fair comparison between methods, placed both task objects and visual distractors in the same configuration across methods by comparing the first frame of the compared camera views for each rollout condition.

E. Robot Hardware & Data Rate

We build Acoustic-Video perception on a bimanual robot platform built from two I2RT YAM arms with linear grippers, controlled at 50 Hz. Each arm has an Intel RealSense D405 wrist camera streaming 640×480 image at 30 FPS. The acoustic image is computed from 2048-sample blocks of the 44.1 kHz, giving an effective beamforming rate of 21.5 Hz. For training, camera images and AV input are both resampled onto the 50 Hz robot control clock, so every policy step sees a temporally aligned acoustic-video observation.

F. Evaluation Metrics

The binary success criteria for each task are described in Sec. VI-A. In addition, we report the task progress rate for tasks in which continuous execution is required for success (3T, 4T, and 5T). For instance, in *Surface Cleaning*, task progress is defined as $S_{\text{progress}} = \frac{B_{\text{erased}}}{B_{\text{initial}}}$ where B is a blue mark and its initial length is 29 cm. Similarly, task progress for *Transparent Film Peeling* and *Sealed-Box Opening* is defined as the normalized length of detached film and cut tape, respectively, relative to their initial total lengths.

VII. RESULTS

The overview of the experiment results is shown in Fig. 5. Across five real-world tasks, acoustic-video perception increases task performance compared to both Vision-Only and ManiWAV-inspired baselines.

Finding 1: Acoustic Video spatially grounds vibroacoustic events to visually relevant interactions. AV-Attention improves task performance by helping the policy attend to task-relevant visual features. In Cable Threading, it increases the success rate to 0.70 (Vision-Only: 0.50, ManiWAV-Inspired: 0.55). The primary failure is exit-detection ambiguity (Fig. 6, *Cable Threading*), causing premature grasping or continued over-pushing. Vision-Only diffuses attention across the background during cable exit, whereas AV-Attention concentrates it on the conduit aperture and the emerging cable (Fig. 7). In Transparent Film Peeling, AV-Attention improves the progress rate to 0.75 (Vision-Only: 0.51, ManiWAV-Inspired: 0.55). Baselines often miss or slip grasps due to unstable attempts, show a lack of

retry behavior. They struggle to attend to visual features of transparent film, while AV-Attention weights the smartphone body and subtle cues such as specular film edges, even under visual distractors (Fig. 7, video figure). In Sealed-Box Opening, baselines (progress 0.29 and 0.11) frequently misaim toward the middle of the seam, yielding partial cuts, whereas AV-Attention shows an explicit aiming stage and more complete cuts (0.55). These suggest that AV-Attention benefits precision tasks where visual features matter and yields more consistent behavior under visual clutter.

Finding 2: Acoustic Video reveals contact state and task progress that vision alone cannot reliably observe.

AV-History excels when the task-relevant state is hard to see. In Surface Cleaning, specular glare obscures the target mark and baselines terminate prematurely after incomplete cleaning (Fig. 6, *Surface Cleaning*). AV-History explicitly tracks cleaned coverage from acoustic video, increasing the success rate to 0.80 (Vision-Only: 0.25, ManiWAV-Inspired: 0.25). In Plastic Grinding, the contact is occluded by the tool in the overhead view. AV-History leverages the acoustic cue at the grinding bit and reaches 0.55 (Vision-Only: 0.40, ManiWAV-Inspired: 0.20). It is likewise robust to occlusion in Cable Threading, reaching 0.75 (Vision-Only: 0.50, ManiWAV-Inspired: 0.55).

Finding 3: Attention and History address complementary perceptual limitations. AV-Attention helps most when visual evidence is present but hard to identify, whereas AV-History helps most when the contact state itself is hard to observe. In Surface Cleaning, guided attention improves mark detection despite the reflective surface (progress 0.53 vs. 0.38 for baselines) yet still terminates prematurely, while AV-History reaches 0.77 by tracking coverage directly. Sealed-Box Opening shows the reverse: the seam is visible but must be precisely located, and AV-Attention shows the explicit aiming stage and yields more complete cuts (0.55) than AV-History (0.33). Transparent Film Peeling shows a similar trade-off under visual distractors. AV-History occasionally fails to hold the phone body when distractors are present, whereas AV-Attention behaves more consistently (0.75 vs.

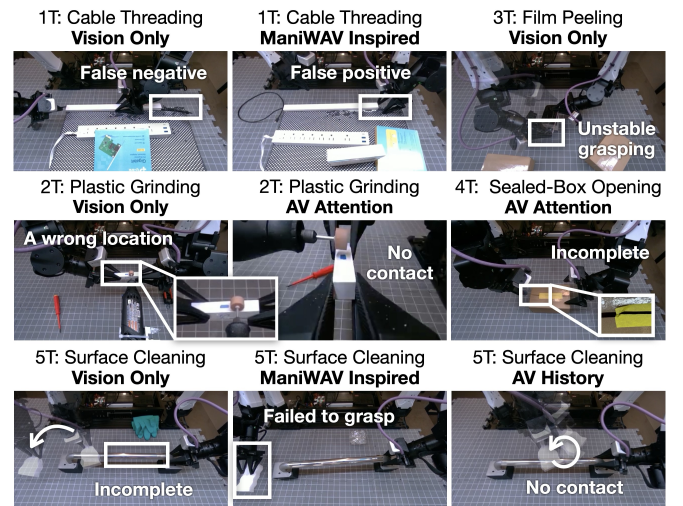


Fig. 6. Failure modes observed during evaluation.

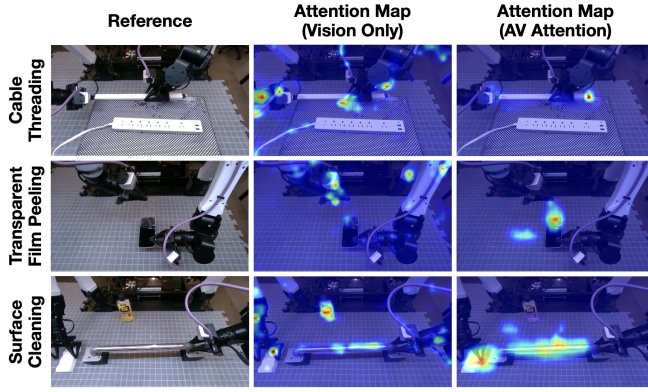


Fig. 7. Action-query-to-image-patch attention map overlaid on the input frame. Middle: Without AV-Attention, the policy spreads attention over background clutter (See *Surface Cleaning*). Right: Models with AV-Attention better attend to task-relevant pixels, e.g., attending to the USB plug tip in *Cable Threading* or even to transparent materials better in *Transparent Film Peeling*.

0.56). This suggests that AV-Attention re-weights what the pixels already contain, while AV-History supplies contact information the pixels do not.

Finding 4: Spatial grounding may make acoustic features more robust than ungrounded spectrogram representations. The ManiWAV-Inspired baseline often underperforms Vision-Only, most notably on Plastic Grinding and Sealed-Box Opening. We attribute this to its ungrounded audio representation. A mono-channel spectrogram does not associate a sound with a location, so the policy may overfit to acoustic patterns that correlate with the demonstrated actions, making it sensitive to environmental noise and distracting it from the fine-grained visual cues these tasks require. By contrast, Acoustic Video gives the policy a signal that is both informative and spatially consistent, which may explain why our AV methods improve over Vision-Only where the ManiWAV-Inspired baseline does not.

Other failure modes. Beyond those above, we observed several minor failures (Fig. 6). AV-Attention occasionally leaves a sub-millimeter strip of tape uncut in Sealed-Box Opening, which still prevents the flap from opening despite an otherwise complete cut. AV-History sometimes loses contact during Surface Cleaning and, although it correctly recognizes the failure and retries cleaning the same location, it cannot recover from the failure.

VIII. LIMITATION & FUTURE WORK

Although we applied acoustic filters to remove ambient noise (e.g., mechanical vibration, AC-driven lighting, etc., whose fundamental frequencies span mostly < 100 Hz), our pipeline does not consider other types of audio distractors with harmonic frequencies (e.g., music, human speech, etc). Because our target beamforming is at 10 kHz (Sec. IV-B), such distractors are partially attenuated; however, their harmonics can still produce spurious activations in the acoustic video. Addressing this robustly is a promising direction for future work, for instance, through spatial masking of fixed distractor locations or using a source classifier.

We currently supervise attention over the entire remaining future horizon. Although we found this more effective than

supervising only the current acoustic-video frame, the entire future horizon may be unnecessary. Supervision with a fixed-time or contextual-level window needs further exploration.

As defined in Sec. III, our current implementation around the AV-History approach performs no dynamic tracking. Extending it to handle moving objects remains future work.

Some commercial robots already ship with built-in microphone arrays (e.g., Unitree G1, R1). However, such arrays have been used for sound-source separation for human-robot voice interaction (i.e., isolating a human speaker in a noisy environment [55], [56]), and none have been applied for visually grounded acoustic sensing and manipulation. Extending our approach to these platforms is a promising direction, as the sensing hardware is already in place.

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